

In [31]:

```
#Day 3
#Strings
str1 = "Python"
print(type(str1) , str1)

str2 = 'This is a string'
print(type(str2) , str2)

str3 = '''This is also a string,
        but can be mutiline'''
print(type(str3) , str3)

<class 'str'> Python
<class 'str'> This is a string
<class 'str'> This is also a string,
        but can be mutiline
```

In [28]:

```
for i in range(0,len(str1)): #i --> 0 to length of str1 -1 --> 6-1 --> 5
    print(str1[i])
```

P
y
t
h
o
n

In [30]:

```
print(str1[14])
```

```
-----
-
IndexError                                Traceback (most recent call las
t)
<ipython-input-30-7d7ce9ed3b27> in <module>
----> 1 print(str1[14])

IndexError: string index out of range
```

In [37]:

```
# [] slicing a string in python
print(str1[:])
print(str1[0:3])
print(str1[3:])
print(str1[:5])
```

Python
Pyt
hon
Pytho

In [38]:

```
print(str1[::-1])
```

nohtyP

In [46]:

```
#python : 0 to 5 (L to R), -1 to -5 (R to L)
print(str1[:])
print(str1[-1])
print(str1[-3:-1])
print(str1[-4:-1])
```

Python

n

ho

tho

In [47]:

```
str4 = "immutable"
str4[1] = "u"
print(str4)
```

```
-----
-
TypeError                                Traceback (most recent call las
t)
<ipython-input-47-df581269704b> in <module>
      1 str4 = "immutable"
----> 2 str4[1] = "u"
      3 print(str4)
```

TypeError: 'str' object does not support item assignment

In [48]:

```
str4 = "updated immutable"
print(str4)
```

updated immutable

In [49]:

```
del(str4[2])
```

```
-----
-
TypeError                                Traceback (most recent call las
t)
<ipython-input-49-5e4c4902093f> in <module>
----> 1 del(str4[2])
```

TypeError: 'str' object doesn't support item deletion

In [52]:

```
list = [1,2,3,4,5]  #mutable  
list[1] = 45  
print(list)
```

[1, 45, 3, 4, 5]

In [53]:

```
del (list[3])  
print(list)
```

[1, 45, 3, 5]

In [55]:

```
str5 = str1 + " " + str2  
print(str5)
```

Python This is a string

In [56]:

```
str6 = str1 * 3  
print(str6)
```

PythonPythonPython

In [66]:

```
print(str2)  
print("is" in str2)  
print("ini" not in str2)
```

This is a string

True

True

In [67]:

```
print("D:\\Python")
```

D:\Python

In [68]:

```
print(r"D:\\Python")
```

D:\\Python

In [71]:

```
print("\\")
```

\

In [72]:

```
print("The String is %s"%str1)
```

The String is Python

In [75]:

```
str7 = "123.56"  
print("The String is %s"%str7)
```

The String is 123.56

In [88]:

```
print("This is aString")  
print("This is a\tString")  
print("This is a \v String")  
print("\110\111\112\113\114")  
print("\x48")
```

This is aString
This is a String
This is a   String
HIJKL
H

In [90]:

```
print("Hello {}, How are you".format("Rajesh"))
```

Hello Rajesh, How are you

In [91]:

```
print("Hello {} and {}".format("Rohan","Sohan"))
```

Hello Rohan and Sohan

In [92]:

```
name1 = input("Enter the name")  
name2 = input("Enter the name")  
print("Hello {} and {}".format(name1,name2))
```

Enter the nameJoe
Enter the nameRaj
Hello Joe and Raj

In [5]:

```
num1 = 10  
float1 = 23.45  
name1 = "Rohan"  
print("Number = %d\nFloat = %f\nName = %s"%(num1,float1,name1))
```

Number = 10
Float = 23.450000
Name = Rohan

In [12]:

```
str8 = "heLLo!! PyTHon"
print(str8.capitalize()) # converts the first alphabet to uppercase
```

Hello!! python

In [14]:

```
print(str8)
print(str8.center(20))
print(str8.center(20, "#"))
```

```
heLLo!! PyTHon
    heLLo!! PyTHon
####heLLo!! PyTHon###
```

In [20]:

```
str9 = "ab bc ab bc ac ca ba ca cb ab ab cc ccc ababa"
print(len(str9))
print(str9.count("ab"))
#print(str9.count("b"))
#print(str9.count("c"))
#print(str9.count(" "))
print(str9.count("ab", 2))

print(str9.count("ab", 2, len(str9)-3))
```

```
45
6
5
4
```

In [53]:

```
print(str9)
print(str9.startswith("aba"))
print(str9.startswith("ab"))
print(str9[3:len(str9)])
print(str9.startswith("bc", 3, len(str9)))
print(str9.startswith("ab", 3, len(str9)))
```

```
ab bc ab bc ac ca ba ca cb ab ab cc ccc ababa
False
True
bc ab bc ac ca ba ca cb ab ab cc ccc ababa
True
False
```

In [51]:

```
print(str9.endswith("aba"))
print(str9.endswith("ab"))
print(str9[0:len(str9)-3])
print(str9.endswith("aba",0,len(str9)-3))
print(str9.endswith("ab",0,len(str9)-3))
```

```
ab bc ab bc ac ca ba ca cb ab ab cc ccc ababa
False
True
bc ab bc ac ca ba ca cb ab ab cc ccc ababa
False
False
=====
True
False
ab bc ab bc ac ca ba ca cb ab ab cc ccc ab
False
True
```

In [36]:

```
print(str9.index("ab"))
print(str9.index("bc"))
#print(str9.index("bc",4,8)) #ValueError: substring not found
print(str9.find("bc",4,8))
```

```
0
3
-1
```

In [42]:

```
str10 = "HELLO$%@"
print(str10.isupper())
```

True

In [43]:

```
print(str10.lower())
```

hello\$%@!

In [46]:

```
print(str10.replace("L","T"))
print(str10.replace("L","T",1))
```

```
HETTO$%@!
HETLO$%@!
```

In [49]:

```
str11 = "Rohan#Mohan#Raj#Sahil#Joe";
list = str11.split("#")
print(list)
print(list[0])
```

```
['Rohan', 'Mohan', 'Raj', 'Sahil', 'Joe']
Rohan
```

In [57]:

```
print(str8)
print(str8.swapcase())
print(str8.title())
```

```
hello!! PyTHon
HELLO!! pYthON
Hello!! Python
```

In [58]:

```
str12 = '''This is a \n multiline \n string'''
print(str12)
```

```
This is a
multiline
string
```

In [61]:

```
print(str12.splitlines())
str13 = str12.splitlines()
print("".join(str13))
```

```
['This is a ', ' multiline ', ' string']
This is a  multiline  string
```

In [77]:

```
str14 = str10.center(40,"*")
print(str14)
print(str14.rstrip('*'))
print(str14.lstrip('*'))
```

```
*****HELLO$%@!*****
*****HELLO$%@!
HELLO$%@!*****
```

```
num = "345678"
```

In [83]:

```
num = "345678"  
print(num)  
print(num.zfill(10))  
print(num.zfill(2))  
print(num.zfill(6))  
print(num.zfill(7))
```

```
345678  
0000345678  
345678  
345678  
0345678
```

In [85]:

```
num = "+234"  
print(num)  
print(num.zfill(10))  
  
num = "-234"  
print(num)  
print(num.zfill(10))
```

```
+234  
+000000234  
-234  
-000000234
```

In []: